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Adaptive Cartesian space control of robotic manipulators: A concurrent learning based approach ☆

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ABSTRACT

This work introduces a concurrent learning-based adaptive control design for end-effector tracking and the corresponding stability analysis for robotic manipulators. The presented controller is developed directly in Cartesian space, thereby removing the necessity for inverse kinematics calculations at the position level. The designed adaptive controller ensures global exponential tracking of the end-effector in Cartesian space. Moreover, the developed controller assures globally exponential convergence of uncertain dynamical parameters to their actual values without demanding persistence excitation conditions via a combination of a standard gradient-based adaptation with a novel integral concurrent learning component. The developed integral concurrent learning part operates both real-time output data and the most informative historical data gathered by employing the singular-value maximization algorithm (SVMA) to reduce the size of memory allocation. Lyapunov-based arguments are applied to ensure the exponential stability of the closed-loop system. Numerical studies are performed to depict the feasibility and performance of the proposed design.

1. Introduction

The control objective for robotic devices is, mostly, defined in the robots' operational space, also known by the name of Cartesian space, end-effector space, or task space. Therefore the actual reference trajectory that the robot has to follow is designed in Cartesian space. However, the dynamics of robotic manipulators (RMs) are generally expressed in joint space and also due to the placement of actuators, the control input signal is defined in the joint space. Methodologies relying on transforming the control objective into the joint space, designing the control input, and then re-transforming the result back to Cartesian space, demand the computation of inverse and direct kinematic in every control cycle. In this approach, the need for inverse kinematics calculations at the position level might become problematic. Researchers have proposed designing direct Cartesian space controllers to avoid the computational burden and also to relax the numerical singularity issues [1].

Direct Cartesian space control is also advantageous when controlling kinematically redundant RMs whose dimension of joint space *i.e.*, number of degrees of freedom (DoF) exceeds the dimension of their operational space [2]. Specifically for kinematically redundant RMs, while tracking a reference Cartesian space trajectory is the primary objective, the redundant DoF results in a movement in the null-space of the Jacobian matrix (*i.e.*, self-motion) which might be used to perform secondary tasks without interfering the primary task. Self-motion can be used to improve the manipulability [3,4], to assure the joints remain within their

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respective limits [5], for grasping objects [6], for obstacle avoidance [7,8], etc. Some line of relevant past research was devoted to control of redundant RMs. In [9,10], feedback linearization and computed torque controllers were proposed for redundant robotic arms to provide tracking in Cartesian space. However, the controllers [9,10] needed accurate knowledge of robot dynamics.

All of the above listed adaptive controllers preferred gradient-type adaptations thus parametric uncertainties are only compensated for and identification was not aimed. Some lines of adaptive control focused on tracking control while simultaneously learning the actual values of uncertain dynamic model parameters. Depending on the type of the update law, they can be classified as composite adaptive controllers, concurrent learning-based adaptive controllers and composite learning adaptive controllers [11]. Composite adaptive controllers primarily rely on a single-channel regressor that utilizes only instantaneous data, necessitating the satisfaction of the persistent excitation (PE) condition [12]. In contrast, employing multi-channel regressors, concurrent learning and composite learning adaptive controllers were presented recently for demanding interval excitation condition instead of PE condition. The concurrent learning-based approach incorporates a discrete data regressor capable of processing both historical and real-time data concurrently within a predefined memory allocation size [13–17]. The selection of the most informative historical data is facilitated through SVMA [18]. The composite learning-based controller [19–29] integrates a prediction error derived from the filtered regression function and input signal over a specified finite interval. This integration ensures the prediction error encompasses past and real-time data within a finite time frame. In [19–23], model reference adaptive controllers were developed to identify the actual values of the uncertain parameters. In [24,25], an interval excitation condition was introduced where the filtered regressor matrix is assumed to be excited over $[0, T]$ for some known constant $T \in \mathbb{R}_{>0}$. In [27–29], the filtered regressor matrix was assumed to be excited over a moving-time window. In [26], a least squares modulated composite learning adaptive controller is presented to guarantee exponential convergence in both tracking error and parameter estimation error. The control design in [26] provides an exponential decay rate that is independent of the excitation level across various regressor channels, leading to a more uniform exponential convergence in the accuracy of parameter estimation and trajectory tracking. While the interval excitation condition is improvements over the persistent excitation condition, the controllers in [13–16] require a higher-order derivative of the state to be available for the implementation. The controllers in [19,27,28,30] necessitate the mass matrix to be known precisely. Recently, an integral concurrent learning adaptive controller was designed in [17] which uses historical data and output data concurrently to identify the actual values of uncertain model parameters without requiring higher-order state derivatives. Among these works, only [17,20,24–26] designed controllers (concurrent learning controller [17] and composite learning controller [20,24–26]) for RMs where their joints are ensured to track given reference trajectories. The controller presented in [26] explicitly provides uniform exponential convergence in joint space while also implicitly ensuring exponential convergence in Cartesian space; in addition, the control implementation for [26] necessitates the use of inverse kinematics calculations at the position level, a technique often viewed as challenging.

In this study, we aimed to extend the aforementioned literature for the end-effector tracking (*i.e.*, Cartesian space) control of RMs while avoiding the need for the position level inverse kinematics calculations. Specifically, we have extended the concurrent learning-based adaptive controller to the Cartesian space control of RMs. The control objective is reformulated in Cartesian space where the actual reference trajectory is formed, while the controller formulation is designed directly in joint space where the actuators are placed. This methodology enabled the removal of the computational burden and singularity issues related to the inverse kinematics calculation. The correlation between the Cartesian and joint space of the RMs is established via an extended Jacobian formulation. The overall control formulation ensures exponential end-effector tracking while ensuring perfect parameter estimation, requiring only interval excitation. Numerical studies performed on a two-link planar RM are also presented to depict the performance of the proposed technique which is also compared with the controller in [26] and a classical gradient-based adaptive controller.

2. Kinematic & dynamic models

An n DoF RM operating on an m-dimensional Cartesian space is considered to have the following position level forward kinematics [31]

$$x = f(q), \tag{1}$$

where $x \in \mathbb{R}^m$ represents the generalized position of the end-effector, $q \in \mathbb{R}^n$ is joint position vector, and $f : \mathbb{R}^n \to \mathbb{R}^m$ symbolizes the direct kinematics with $n > m$. The velocity kinematics is modeled as

$$\dot{x} = J\dot{q}, \tag{2}$$

where $\dot{x} \in \mathbb{R}^m$ is the generalized velocity of the end-effector, $\dot{q} \in \mathbb{R}^n$ expresses joint velocity vector, and $J : \mathbb{R}^n \to \mathbb{R}^{m\times n}$ represents the Jacobian matrix formulated as $J \triangleq \frac{\partial f}{\partial q}$.

The dynamics of the RM is presented as [32]

$$M(q)\ddot{q} + V_m(q,\dot{q})\dot{q} + G(q) + F_d\dot{q} = \tau, \tag{3}$$

where $\ddot{q} \in \mathbb{R}^n$ represents the joint acceleration vector, $M : \mathbb{R}^n \to \mathbb{R}^{n\times n}$ denotes the mass/inertia matrix, $V_m : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^{n\times n}$ defines the centripetal-Coriolis matrix, $G : \mathbb{R}^n \to \mathbb{R}^n$ models gravitational effects, $F_d \in \mathbb{R}^{n\times n}$ is the friction matrix, and $\tau \in \mathbb{R}^n$ symbolizes the control input torque.

Mathematical models given above are considered to satisfy the following standard modeling properties.

Assumption 1. The subsequent development requires all kinematic singularities to be avoided *a priori*.

Assumption 2. The kinematic terms in (1), (2) and the dynamic modeling terms in (3) along with first-order partial derivatives are uniformly bounded in time provided their arguments are bounded [33].

Property 1. *The mass matrix M is positive definite, symmetric, and agrees with the subsequent inequality [32]*

$$m_1 \|\xi\|^2 \leq \xi^T M(q)\xi \leq m_2 \|\xi\|^2, \ \forall \xi \in \mathbb{R}^n, \tag{4}$$

where m_1, $m_2 \in \mathbb{R}_{>0}$ are known bounding constants and $\|\cdot\|$ defines the standard Euclidean norm.

Property 2. *The time derivative of M and V_m fulfill the subsequent property [32]*

$$\xi^T \left(\dot{M}(q) - 2V_m(q,\dot{q})\right)\xi = 0, \ \forall \xi \in \mathbb{R}^n. \tag{5}$$

Property 3. *The dynamic modeling terms in (3) can be linearly parameterized as follows*

$$W(q,\dot{q},\ddot{q})\theta = M(q)\ddot{q} + V_m(q,\dot{q})\dot{q} + G(q) + F_d\dot{q}, \tag{6}$$

where $W(q,\dot{q},\ddot{q}) \in \mathbb{R}^{n\times p}$ is the regression-matrix and $\theta \in \mathbb{R}^p$ symbolizes a vector containing uncertain constant terms that depends on physical properties of the RM.

3. Error system development & control design

The aim of the control design is to assure that the generalized end-effector position of the RM tracks a given reference trajectory $x_d(t) \in \mathbb{R}^m$ despite the dynamical terms in (3) being subject to parametric uncertainties. In addition to the end-effector tracking objective, the uncertain dynamical model parameters are aimed to be adaptively identified (*i.e.*, system identification). Since the number DoF, n, is considered to be bigger than the dimension of the end-effector space, m, then the RM has $n-m$ extra DoF. And in addition to end-effector tracking and system identification objectives, the extra DoF is aimed to be used for achieving secondary objectives. Ensuring the stability of the closed-loop dynamics is also aimed. In achieving these objectives, the subsequent control development requires joint position and velocity measurements to be available and the direct kinematics of the RM along with the Jacobian matrix is known.

To quantify the tracking objective, end-effector/Cartesian space tracking error, shown with $e \in \mathbb{R}^m$, is defined as

$$e \triangleq x_d - x, \tag{7}$$

where the reference trajectory is assumed to be designed such that x_d, $\dot{x}_d$, $\ddot{x}_d$ are bounded functions of time.

To make use of the extra DoF, the sub-task function vector, shown with $y_{st}(q) \in \mathbb{R}^{n-m}$ is introduced. The entries of $y_{st}(q)$ shown with $y_{st,i}(q) \in \mathbb{R}$, $i = 1,\ldots,n-m$ denote individual sub-task functions. It is emphasized that the end-effector tracking objective has the highest priority and the sub-task functions are pursued afterward (*i.e.*, they are secondary objectives.). The dynamics of the sub-task function are obtained as

$$\dot{y}_{st} = J_{st}\dot{q}, \tag{8}$$

where $J_{st}(q) \in \mathbb{R}^{(n-m)\times n}$ is a Jacobian-like function, formulated as

$$J_{st} \triangleq \frac{\partial y_{st}}{\partial q}. \tag{9}$$

Remark 1. It is emphasized that detailed consideration is needed when developing the sub-task functions so they do not interfere with the main task. Mathematically, the rows of J_{st} must not be in the range space of J.

To quantify both end-effector tracking and secondary objectives, a combined error function, represented with $\bar{e}(t) \in \mathbb{R}^n$, is introduced as

$$\bar{e} \triangleq \bar{x}_d - \bar{x}, \tag{10}$$

where $\bar{x} \triangleq \begin{bmatrix} x^T & y_{st}^T \end{bmatrix}^T \in \mathbb{R}^n$ and $\bar{x}_d \triangleq \begin{bmatrix} x_d^T & y_d^T \end{bmatrix}^T \in \mathbb{R}^n$ with $y_d \in \mathbb{R}^{n-m}$ including the constant reference values of the sub-task functions y_{st}.

To tie the end-effector tracking error to the joint space, an auxiliary tracking error-like vector, represented with $r \in \mathbb{R}^n$, is introduced as

$$r \triangleq \bar{J}^{-1}\left(\dot{\bar{x}}_d + \alpha\bar{e}\right) - \dot{q}, \tag{11}$$

where $\alpha \in \mathbb{R}^{n\times n}$ is a constant, diagonal, positive definite gain matrix and $\bar{J}(q) \in \mathbb{R}^{n\times n}$ is formulated as

$$\bar{J} \triangleq \begin{bmatrix} J \\ J_{st} \end{bmatrix}. \tag{12}$$

It is noted that in view of Assumption 1 and Remark 1, $\bar{J}$ is full rank and its inverse exists for every q.

The dynamics of $\bar{e}$ is obtained by multiplying (11) with $\bar{J}$ and then using (2), (8), and the time derivative of (10) to yield

$$\dot{\bar{e}} = -\alpha\bar{e} + \bar{J}r. \tag{13}$$

The open-loop dynamics for r can be obtained by multiplying the time derivative of (11) with M then utilizing (3) and (11) to reach

$$M\dot{r} = -V_m r + Y\theta - \tau, \tag{14}$$

where $Y(q, \dot{q}, \bar{x}_d, \dot{\bar{x}}_d, \ddot{\bar{x}}_d) \in \mathbb{R}^{n\times p}$ is obtained from

$$Y\theta = M\frac{d}{dt}\left(\bar{J}^{-1}\left(\dot{\bar{x}}_d + \alpha\bar{e}\right)\right) + V_m\bar{J}^{-1}\left(\dot{\bar{x}}_d + \alpha\bar{e}\right) + G + F_d\dot{q}, \tag{15}$$

in which θ was previously introduced in Property 3.

The control signal is developed depending on the subsequent stability analysis and is designed as

$$\tau = Kr + \bar{J}^T\bar{e} + Y\hat{\theta} \tag{16}$$

where $K \in \mathbb{R}^{n\times n}$ defines a positive definite diagonal control gain matrix and $\hat{\theta}(t) \in \mathbb{R}^p$ is the estimate of θ. The term $\hat{\theta}(t)$ is designed to be adjusted in agreement with the following adaptive law

$$\dot{\hat{\theta}} = \Gamma Y^T r + k\Gamma\sum_{i=1}^{N}\mathcal{Y}_i^T\left(\mathcal{U}_i - \mathcal{Y}_i\hat{\theta}\right). \tag{17}$$

In (17), $\Gamma \in \mathbb{R}^{p\times p}$ defines a constant positive definite diagonal adaptation gain matrix, $k \in \mathbb{R}_{>0}$ is a constant, $\mathcal{Y}_i \triangleq \mathcal{Y}(t_i)$ and $\mathcal{U}_i \triangleq \mathcal{U}(t_i)$, with t_i being a time point that is between the initial time and the current time, *i.e.*, $t_i \in [t_0, t]$. The terms $\mathcal{U}(t) \in \mathbb{R}^n$ and $\mathcal{Y}(t) \in \mathbb{R}^{n\times p}$ are presented as

$$\mathcal{U} \triangleq \int_{\max\{t-\Delta t, t_0\}}^{t}\tau(\sigma)\,d\sigma, \tag{18}$$

$$\mathcal{Y} \triangleq W_1(q(t), \dot{q}(t), q(t-\Delta t), \dot{q}(t-\Delta t)) + \int_{\max\{t-\Delta t, t_0\}}^{t} W_2(q(\sigma), \dot{q}(\sigma))\,d\sigma, \tag{19}$$

where $\Delta t \in \mathbb{R}_{>0}$ is a constant that is used to define the window of integration and $W_1(t)$, $W_2(t) \in \mathbb{R}^{n\times p}$ are obtained from

$$W_1\theta \triangleq M(q(t))\dot{q}(t) + M(q(t-\Delta t))\dot{q}(t-\Delta t), \tag{20}$$

$$W_2\theta \triangleq -\dot{M}(q(t))\dot{q}(t) + V_m(q(t), \dot{q}(t))\dot{q}(t) + G(q(t)) + F_d\dot{q}(t). \tag{21}$$

Via utilizing (18)–(21), the expression in (17) can be represented as

$$\dot{\hat{\theta}} = \Gamma Y^T r + k\Gamma\sum_{i=1}^{N}\mathcal{Y}_i^T\mathcal{Y}_i\tilde{\theta}, \tag{22}$$

where $\tilde{\theta}(t) \in \mathbb{R}^p$ is the error between actual and estimation of uncertain parameters, expressed as

$$\tilde{\theta} \triangleq \theta - \hat{\theta}. \tag{23}$$

Substituting (16) into (14) and utilizing (23), the closed-loop dynamics for r is obtained as

$$M\dot{r} = -V_m r - Kr - \bar{J}^T\bar{e} + Y\tilde{\theta}. \tag{24}$$

4. Stability analysis

Two cases are to be investigated depending on the minimum eigenvalue of $\sum_{i=1}^{N}\mathcal{Y}_i^T\mathcal{Y}_i$ being compared with an *a priori* determined constant, represented with $\lambda_{\mathcal{Y}} \in \mathbb{R}_{>0}$, for the history stack to be considered sufficiently rich. For the first case, the history stack is considered to be initially not rich enough in the sense that $\lambda_{\min}\left\{\sum_{i=1}^{N}\mathcal{Y}_i^T\mathcal{Y}_i\right\} \ngeq \lambda_{\mathcal{Y}}$ for some initial time period $\forall t \in [t_0, T)$. For the second case, the history stack becomes rich enough to meet $\lambda_{\min}\left\{\sum_{i=1}^{N}\mathcal{Y}_i^T\mathcal{Y}_i\right\} \geq \lambda_{\mathcal{Y}}$ during the time period $\forall t \in [T, +\infty)$. During the first time period, due to the collected data being limited the minimum eigenvalue condition is not satisfied; however, as new data is added the condition is expected to be satisfied.

Theorem 1. *The controller given in (16) in conjunction with the novel adaptation law designed in (17) ensure GES of the end-effector tracking, sub-task, and parameter estimation errors in the sense that*

$$\|s(t)\| \leq \sqrt{\frac{\bar{\lambda}_V}{\underline{\lambda}_V}}\,\|s(T)\|\exp\left(-\frac{\kappa}{2\bar{\lambda}_V}(t-T)\right), \tag{25}$$

$\forall\, t \in [T, +\infty)$, *where* $s(t) \triangleq \begin{bmatrix}\bar{e}^T & r^T & \tilde{\theta}^T\end{bmatrix} \in \mathbb{R}^{2n+p}$, $\bar{\lambda}_V$, $\underline{\lambda}_V$, $\kappa \in \mathbb{R}_{>0}$ *are constants.*

Proof. For $D \subset \mathbb{R}^{2n+p}$ being a domain containing the origin, let $V(s,t) : D \times [0,+\infty) \to \mathbb{R}$ be a positive definite, continuously differentiable, and radially unbounded Lyapunov function candidate presented as

$$V \triangleq \frac{1}{2}\bar{e}^T\bar{e} + \frac{1}{2}r^T M(q) r + \frac{1}{2}\tilde{\theta}^T \Gamma^{-1}\tilde{\theta}. \tag{26}$$

The following inequality can be written for the above-given Lyapunov function via

$$\underline{\lambda}_V \|s\|^2 \le V(s) \le \bar{\lambda}_V \|s\|^2, \tag{27}$$

where $\underline{\lambda}_V \triangleq \frac{1}{2}\min\{1, \underline{m}, \underline{\lambda}_\Gamma\}$ and $\bar{\lambda}_V \triangleq \frac{1}{2}\max\{1, \bar{m}, \bar{\lambda}_\Gamma\}$ with $\underline{\lambda}_\Gamma$, $\bar{\lambda}_\Gamma \in \mathbb{R}$ are the minimum and the maximum eigenvalues of Γ^{-1}, respectively.

The time derivative of (26) is obtained as

$$\dot{V} = \bar{e}^T\dot{\bar{e}} + r^T M\dot{r} + \frac{1}{2}r^T\dot{M}r + \tilde{\theta}^T\Gamma^{-1}\dot{\tilde{\theta}}. \tag{28}$$

Substituting (13) for the dynamics of $\bar{e}$, (24) for the dynamics of r, (22) along with the time derivative of (23) for the dynamics of $\tilde{\theta}$, and then using Property 2 along with straightforward simplifications results in

$$\dot{V} = -\bar{e}^T\alpha\bar{e} - r^T K r - k\tilde{\theta}^T\left(\sum_{i=1}^{N}\mathcal{Y}_i^T\mathcal{Y}_i\right)\tilde{\theta}. \tag{29}$$

Since the history stack is assumed to be not rich enough for some initial time period of $t \in [t_0, T)$, then $\sum_{i=1}^{N}\mathcal{Y}_i^T\mathcal{Y}_i$ is positive semi-definite and the right-hand side of (29) is upper bounded as

$$\dot{V} \le -\bar{e}^T\alpha\bar{e} - r^T K r \le -\underline{\lambda}_\alpha\|\bar{e}\|^2 - \underline{\lambda}_K\|r\|^2, \tag{30}$$

where $\underline{\lambda}_\alpha$, $\underline{\lambda}_K \in \mathbb{R}$ are the minimum eigenvalues of α and K, respectively. The expression (30) can further be bounded above as

$$\dot{V} \le -\gamma(s), \ \forall t \in [t_0, T), \tag{31}$$

where $\gamma : \mathbb{R}^{2n} \to \mathbb{R}_{\ge 0}$ is a continuous, negative semi-definite function defined as $\gamma \triangleq \min\{\underline{\lambda}_\alpha, \underline{\lambda}_K\}\left(\|\bar{e}\|^2 + \|r\|^2\right)$. Using Theorem 8.4 of [34], the inequality in (31) can be used to conclude that V and thus s are bounded functions of time. Further analysis can be used to demonstrate the global boundedness of the remaining closed-loop signals. Since the controller of (16) along with the adaptive update law in (17) ensure that entire signals remain bounded, the possibility of finite escape phenomena during $\forall t \in [t_0, T)$ is avoided [35]. Furthermore, following the footsteps of the analysis associated with standard gradient-type adaptive controllers, Barbalat's Lemma [34] can be invoked to prove the asymptotic stability of $\bar{e}$ and r.

The above analysis was for the scenario when the history stack is insufficiently rich to fulfill requirement $\lambda_{\min}\left\{\sum_{i=1}^{N}\mathcal{Y}_i^T\mathcal{Y}_i\right\} \ge \lambda_{\mathcal{Y}}$ during $\forall t \in [t_0, T)$. After some time T, the history stack becomes rich enough to meet $\lambda_{\min}\left\{\sum_{i=1}^{N}\mathcal{Y}_i^T\mathcal{Y}_i\right\} \ge \lambda_{\mathcal{Y}}$. So $\forall t \ge T$, $\sum_{i=1}^{N}\mathcal{Y}_i^T\mathcal{Y}_i$ is positive definite and the right hand side of (29) can now be upper bounded as

$$\dot{V} \le -\underline{\lambda}_\alpha\|\bar{e}\|^2 - \underline{\lambda}_K\|r\|^2 - k\lambda_{\mathcal{Y}}\|\tilde{\theta}\|^2. \tag{32}$$

In view of (27), the right hand side of (32) can further be upper bounded as

$$\dot{V} \le -\frac{\kappa}{\bar{\lambda}_V}V, \ \forall t \in [T, \infty), \tag{33}$$

where $\kappa \triangleq \min\{\underline{\lambda}_\alpha, \underline{\lambda}_K, k\lambda_{\mathcal{Y}}\}$. The above differential inequality can be solved to conclude that

$$V(t) \le V(T)\exp\left(-\frac{\kappa}{\bar{\lambda}_V}(t-T)\right), \ \forall t \in [T, \infty), \tag{34}$$

which, in view of (27), yields the upper bound given in (25). □

5. Numerical results

A numerical simulation is performed to depict the performance of the controller in (16) in conjunction with the adaptation law in (17) for a two-link planar RM. The direct kinematics is given as

$$x = \begin{bmatrix} L_1c_1 + L_2c_{12} \\ L_1s_1 + L_2s_{12} \end{bmatrix},$$

from which the Jacobian matrix can be found as

$$J = \begin{bmatrix} -L_1s_1 - L_2s_{12} & -L_2s_{12} \\ L_1c_1 + L_2c_{12} & L_2c_{12} \end{bmatrix},$$

where s_a, c_a, s_{ab}, c_{ab} express $\sin(q_a)$, $\cos(q_a)$, $\sin(q_a + q_b)$ and $\cos(q_a + q_b)$, respectively. The link lengths of $L_1 = 0.4$m and $L_2 = 0.36$m were chosen. The dynamics of the planar RM is given by

$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} p_1 + 2p_3c_2 & p_2 + p_3c_2 \\ p_2 + p_3c_2 & p_2 \end{bmatrix}\begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{bmatrix}$$

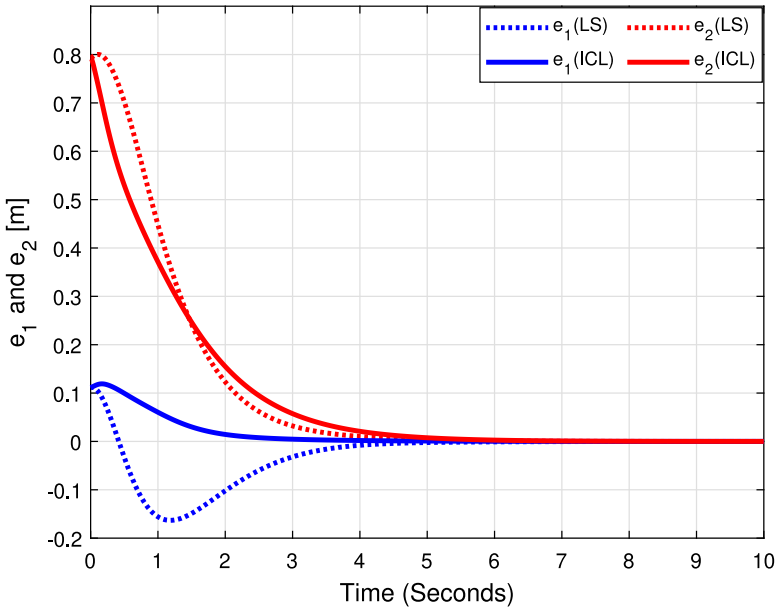

Fig. 1. The Cartesian space tracking error.

$$+\begin{bmatrix} -p_3 s_2 \dot{q}_2 & -p_3 s_2 \left(\dot{q}_1 + \dot{q}_2\right) \\ p_3 s_2 \dot{q}_2 & 0 \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} + \begin{bmatrix} f_{d1}\dot{q}_1 \\ f_{d2}\dot{q}_2 \end{bmatrix},$$

with the model parameters $p_1 = 3.473$, $p_2 = 0.196$, $p_3 = 0.242$ and dynamic friction coefficients $f_{d1} = 5.3$, $f_{d2} = 1.1$. The vector that contains the uncertain model parameters for the planar RM was assembled as $\theta = \begin{bmatrix} p_1 & p_2 & p_3 & f_{d1} & f_{d2} \end{bmatrix}^T$. The reference Cartesian space trajectory was preferred as

$$x_d = \begin{bmatrix} 0.4 + 0.1\cos(t)\left(1 - \exp(-0.5t)\right) \\ 0.4 + 0.1\sin(t)\left(1 - \exp(-0.5t)\right) \end{bmatrix} \text{m},$$

and the robot arm is supposed to be initially at rest at the joint configurations of $q_1(0) = -1.75$rad and $q_2(0) = 1.75$rad. The initial value of the vector $\hat{\theta}$ was selected as 0. Controller and adaptation gains were selected via trial and error and chosen as

$$\alpha = \text{diag}\{1, 1\}, \qquad K = \text{diag}\{10, 10\},$$
$$\Gamma = \text{diag}\{7.5, 7.5, 7.5, 7.5, 7.5\}, \qquad k = 200.$$

The size of the collected data was selected as 50 when implementing SVMA and a fixed step size of 0.001 was chosen for all simulations. The performance of the designed controller against the least square-modulated composite learning controller in [26] is presented to demonstrate the effectiveness of the proposed direct task space control methodology. In the case that the reference trajectory for the end-effector of RM is presented in task space, to implement the controller in [26], it is necessary to map the reference trajectory from Cartesian space to the joint space. Even though the effectiveness of the controller in [26] for joint space tracking has been proven, the exponential decay rate of convergence may not be assured for task space tracking applications, mainly due to the need for mapping position and velocity vectors by nonlinear functions (the forward kinematics function and the Jacobian matrix).

Comparative numerical simulations were conducted using a reference trajectory for the end-effector defined in task space. For the controller in [26], this reference trajectory in task space was transformed into joint space coordinates using the inverse kinematic solution of a 2-link revolute RM. For both the designed controller and the controller [26], the initial conditions, the reference trajectory, control gains α and K, and the adaptation gain Γ are kept identical. Additionally, the controller in [26] meets the interval excitation criteria for $t \in [0, 0.1]$, the learning gain is selected as $k_w = 1$, and the filter term is selected as $\beta = 10$. In all figures and tables, variables pertaining to the designed CLAC controller are labeled with (ICL), while those associated with the least square-modulated composite learning controller in [26] are marked with (LS). Figs. 1–4 present the results of the numerical simulations conducted to compare the performance of the designed controller and LS. Fig. 1 depicts the comparison of position errors in task space between the designed controller and the controller in [26]. The designed controller achieves exponential convergence of the position error in task space, whereas the controller in [26] provides exponential convergence with a slightly bigger envelope. Fig. 2 shows the control input torque for both the designed and the one in [26]. The designed controller is constructed of error dynamics in task space, while the controller in [26] is formed of errors in joint space, leading to non-identical initial errors for each controller. As a result, a direct comparison of their transient responses is not equitable due to the strong influence of these initial errors. Nonetheless, as Fig. 2 indicates, their steady-state responses are nearly identical. Fig. 3 displays the parameter estimation

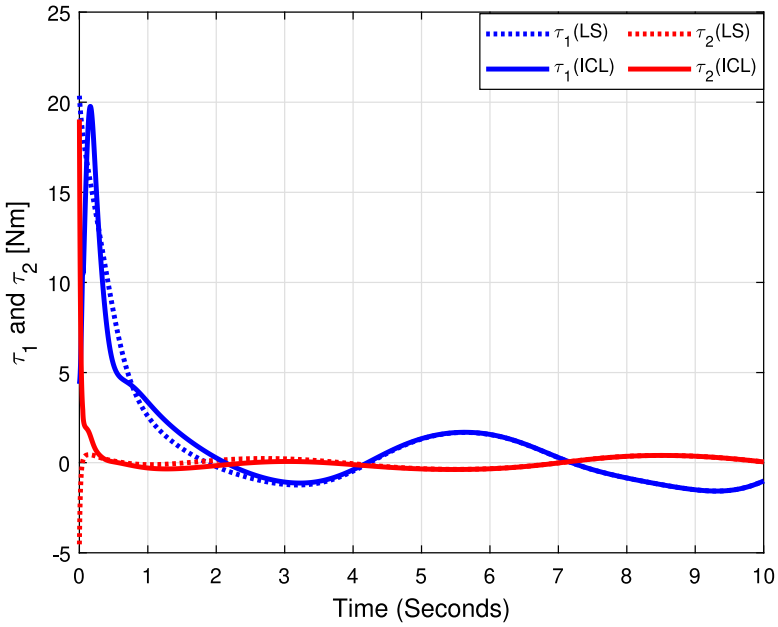

Fig. 2. The control input torque $\tau(t)$.

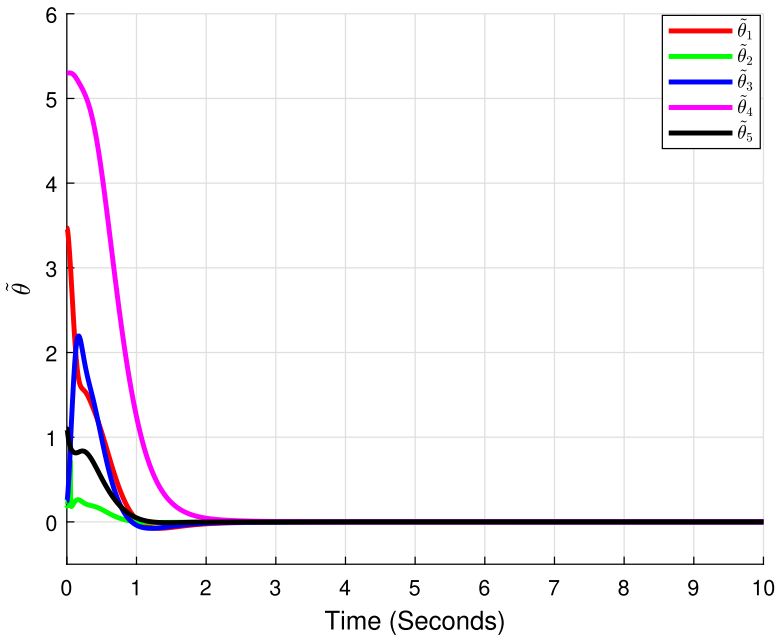

Fig. 3. The entries of the parameter estimation error.

errors for the designed controller, contrasting actual and estimated values of uncertain terms. It demonstrates that the designed controller and its update law facilitate exponential convergence of parameter adaptation to their true values. Fig. 4 compares the identification performance of both controllers, highlighting that the controller in [26] identifies uncertain terms slightly faster than the designed controller.

To compare the identification and tracking performances of the designed controller, the controller in [26], and a classical gradient-based adaptive controller, labeled as CGAC, in steady-state duration, Table 1 is presented. The CGAC is obtained by turning off the concurrent learning component of the adaptive update rule in (17) via setting $k = 0$ while keeping the rest of the design parameters the same. The root mean square (RMS) values of the entries of the tracking error e_1, e_2, the control input torque τ_1, τ_2, and the parameter estimation errors are evaluated and given in Table 1 for the duration of the simulation $t \in [0, 10]$ and after

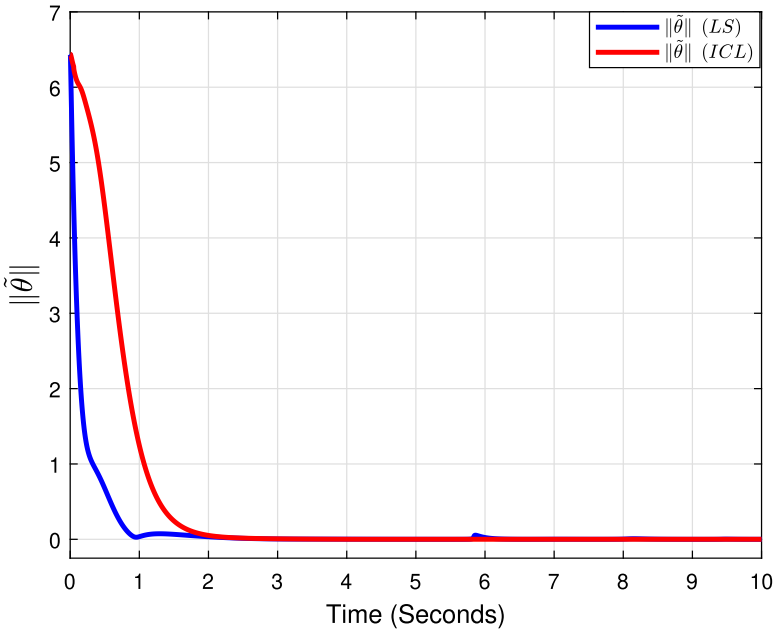

Fig. 4. The norm of the parameter estimation error.

Table 1
RMS Values for ICL, LS, and CGAC.

	ICL $t \in [0, 10]$	LS $t \in [0, 10]$	CGAC $t \in [0, 10]$	ICL $t \in [5, 10]$	LS $t \in [5, 10]$	CGAC $t \in [5, 10]$
e_1	0.0331	0.0588	0.0390	0.0003	0.0007	0.0206
e_2	0.1998	0.2344	0.1906	0.0024	0.0012	0.0265
τ_1	3.2458	3.5189	4.0114	1.2283	1.2245	1.0635
τ_2	0.7088	0.2754	0.7248	0.2966	0.2944	0.2930
p_1	0.4470	0.3149	3.2975	0.0003	0.0002	3.3077
p_2	0.0569	0.0248	1.2714	0.0000	0.0001	0.9925
p_3	0.3708	0.1292	0.7469	0.0005	0.0002	0.4464
f_{d1}	1.2761	0.4204	3.5730	0.0004	0.0003	3.2335
f_{d2}	0.1864	0.0911	0.9125	0.0003	0.0001	0.8611

steady state is reached $t \in [5, 10]$. The initial conditions for position errors in both task space and joint space differ between the designed controller and the controller in [26]; as a result, directly comparing their transient responses might not be an equitable evaluation. Therefore, comparing the performance of the controllers after they have achieved a steady-state response would be advantageous. The data in Table 1 shows that both the designed controller and the controller in [26] exhibit almost the same level of performance in tracking and identification tasks during the time interval $t \in [5, 10]$. Additionally, in the time interval $[5, 10]$, both the designed controller and the controller in [26] outperform CGAC in terms of tracking and identification. Table 2 displays the performance metrics for the designed controller, the controller in [26], and CGAC when subjected to time-varying additive disturbances, specifically $d(t) = \frac{1}{2}\,[\sin(2t) \;\; \cos(0.5t)]^T$. According to Table 2, the controller in [26] achieves better tracking performance compared to both the designed controller and CGAC. Furthermore, the designed controller exhibits better tracking capabilities than CGAC. However, in terms of robustness against additive disturbances for identification tasks, the designed controller outperforms both the controller in [26] and CGAC.

6. Conclusions

The design and correspondent stability analysis of a CLAC for RM in Cartesian space has been presented. The presented adaptation update law used the chosen historical data and the output data simultaneously to complete sufficiently rich data. The SVMA was employed to pick the most informative historical data to fulfill the predefined memory. A Lyapunov-based stability analysis was utilized to verify exponential stability for the end-effector position tracking and the positions of the joints that contribute to the self-motion. The proposed controller ensures exponential tracking error convergence and exact estimation of system parameters, provided that an interval excitation is satisfied. Numerical studies were presented to confirm the performance of the designed controller and its comparison with a standard gradient-type adaptive controller. When compared with the works in [17,20,24–26], the main advantages of the proposed framework are twofold. Firstly, since most robotics applications require the positioning of

Table 2
RMS Values for ICL, LS, and CGAC with disturbances.

	ICL $t \in [0, 10]$	LS $t \in [0, 10]$	CGAC $t \in [0, 10]$	ICL $t \in [5, 10]$	LS $t \in [5, 10]$	CGAC $t \in [5, 10]$
e_1	0.0358	0.0586	0.0399	0.0125	0.0015	0.0233
e_2	0.1977	0.2340	0.1896	0.0082	0.0015	0.0299
τ_1	3.2174	3.4621	4.0208	1.2606	1.3087	1.1283
τ_2	0.7465	0.3833	0.7574	0.4247	0.4264	0.4206
p_1	0.5199	2.5865	3.2936	0.2934	3.2971	3.2721
p_2	0.0598	0.7139	0.9579	0.0211	0.7440	0.5903
p_3	0.4044	2.4728	0.8231	0.1999	2.5599	0.5586
f_{d1}	1.2991	2.6420	3.6216	0.1917	2.5540	3.2980
f_{d2}	0.3687	1.0252	1.0575	0.3098	0.8921	1.0540

the end-effector of the RMs accurately, in this study, tracking is achieved in the Cartesian space where all of the relevant past works aimed at joint space tracking. Secondly, in the case of kinematic redundancy, a systematic approach is proposed to integrate secondary objectives while fulfilling the end-effector tracking objective which is ensured exponentially. Future research will focus on identification of kinematic model parameters, in addition to accurately estimating dynamic model parameters. Another interesting future work could be designing the direct task space version of the controller in [26].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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